

# IoT & Machine Learning course schedule

Jul 13 – Aug 6, 2026 · Lilly Hall of Life Sciences 2102

Class (CA)    Pre-lab (PLA)    Lab (LA)    Project    Special

## WEEK 1 · JUL 13–17

### Day 1 Mon Jul 13

#### CA

9:00–10:45 AM

Introduction (Prof. Martin Jun) 9:00–9:30

ML1-1: Data Analysis – Handling & Plotting

#### PLA · PL0

11:15 AM–2:15 PM

PL0 Colabs 1–3 (Setting up Raspberry Pi)

#### LA · L0

2:15–4:30 PM

L0 Colab 1

### Day 2 Tue Jul 14

#### CA

9:00–10:45 AM

Guest Speaker 9:00–9:50

ML1-2: Data Analysis – Data Indexing

ML1-3: Data Processing

#### PLA · PL1

11:15 AM–2:15 PM

PL1 Colabs 1–2 (Environmental Sensors)

#### ISS check-in

2:30–4:40 PM

Purdue ISS · no lab

### Day 3 Wed Jul 15

#### CA

9:00–10:45 AM

Guest Speaker 9:00–9:50

ML2-1: Signal Processing 1 (FFT)

ML2-2: Signal Processing 2 (STFT)

#### PLA · PL1

11:15 AM–1:15 PM

PL1 Colab 3 (Environmental Sensors)

#### LA · L1

1:15–3:30 PM

L1 Colabs 1–2

#### Seminar

4:00 PM

### Day 4 Thu Jul 16

#### CA

9:00–10:45 AM

ML2-3: Signal Processing 3 (WT)

#### PLA · PL2

11:15 AM–2:15 PM

PL2 Colabs 1–2 (Accelerometer & Signal Processing)

#### LA · L2

2:15–4:30 PM

L2 Colabs 1–3

**Day 5** Fri Jul 17**CA**

9:00–10:45 AM

ML3-1: Feature Extraction

ML3-2: Feature Selection (p-value)

**PLA · PL3**

11:00 AM–12:00 PM

PL3 Colabs 1–2 (Real-Time Communication)

**LA · L3**

1:30–3:15 PM

L3 Colabs 1–3

**MMRL Tour**

3:30 PM–4:30 PM

**WEEK 2 · JUL 20–24****Day 6** Mon Jul 20**CA**

9:00–10:45 AM

ML4-1: Dim. Reduction &amp; Visualization 1 (PCA)

ML4-2: Dim. Reduction &amp; Visualization 2 (t-SNE)

**PLA · PL4**

11:15 AM–2:15 PM

PL4 Colab 1 (Middleware)

**LA · L4**

2:15–3:30 PM

L4 Colabs 1–3

**MMRL Project**

3:30–4:30 PM

Machine Tool Monitoring &amp; Data Visualization · Kick-off

**Day 7** Tue Jul 21**CA**

9:00–10:45 AM

ML5-1: Artificial Neural Network (ANN)

**PLA · PL5**

11:15 AM–2:15 PM

PL5 Colab 1 (MTConnect &amp; HTTP)

**LA · L5**

2:15–3:30 PM

L5 Colabs 1–3

**Industry Tour - Kirby Risk**

3:30 - 5:30 PM

**Day 8** Wed Jul 22**CA**

9:00–10:45 AM

ML6-1: Convolutional Neural Network (CNN)

ML7-1: Grid Search 1 (ANN)

**PLA · PL6**

11:00 AM–12:00 PM

PL6 Colab 1 (SQL)

**Industry Tour - Evonik**

1:30–3:30 PM

**Seminar**

4:00 PM

**Day 9** Thu Jul 23**CA**

9:00–10:45 AM

ML7-2: Grid Search 2 (CNN)

**LA · L6**

11:00 –12:30 PM

L6 Colabs 1–2

**MMRL Project**

1:30–4:30 PM

**Day 10** Fri Jul 24**MMRL Project**

9:00 AM–12:00 PM

**Midterm presentations**

1:30 PM–4:30 PM

MMRL Project midterm presentations

**WEEK 3 · JUL 27–31****Day 11** Mon Jul 27**CA**

9:00–10:45 AM

ML8: Industrial Data Processing

**PLA · PL7**

11:00 AM–12:15 PM

PL7 Colab 1 (Grafana)

MSM dataset briefing for Mini Project

**LA · L7**

1:30–3:30 PM

L7 Colabs 1–2

**MMRL Project**

3:30–4:30 PM

**Day 12** Tue Jul 28**PLA · PL8**

9:00–11:00 AM

PL8 Colabs 1–3 (ML Experiments)

**LA · L8**

11:00–2:30 PM

L8 Colabs 1–2

**PLA · PL9**

2:45–4:30 PM

**Day 13** Wed Jul 29**LA · L9**

9:00–11:30 AM

**MMRL Project**

1:00–3:30 PM

PL9 Colab 1 (Implement ML to Raspberry Pi)

then L9 Colabs 1–2

**Seminar**

4:00 PM

**Day 14** Thu Jul 30**PLA · PL10**

9:00 AM - 11: AM

**LA · L10**

11:15 –4:30 PM

L10 Colabs 1–3

**Day 15** Fri Jul 31**Mini Project + MMRL Project**

9:00 AM–4:30 PM

Finish Mini Project MSM · MMRL Project

**WEEK 4 · AUG 3–6****Day 16** Mon Aug 3**MMRL Project**

9:00 AM–4:30 PM

**Day 17** Tue Aug 4**MMRL Project**

9:00 AM–4:30 PM

**Day 18** Wed Aug 5**MMRL Project · final day**

9:00 AM–12:30 PM

Final refinements & presentation  
preparation**Seminar**

4:00 PM

**Day 19** Thu Aug 6**Final presentations**

9:00 AM–12:00 PM

MMRL Project final presentations ·  
course wrap-up